Summary

Summary

- **The Power Rule**: For any real number n, $\dfrac{d}{dx}x^n = nx^{n-1}$.
- **The Natural Exponential Function**: $\dfrac{d}{dx}e^x = e^x$.
- **The Sine Function**: $\dfrac{d}{dx}\sin(x) = \cos(x)$.
- **The Sum Rule**: If $f(x)$ and $g(x)$ are differentiable functions and c is a constant, then:
  - $\dfrac{d}{dx}\left(f(x) + g(x)\right) = f'(x) + g'(x)$,
  - $\dfrac{d}{dx}\left(f(x) - g(x)\right) = f'(x) - g'(x)$,
  - $\dfrac{d}{dx}\left(c \cdot f(x)\right) = c \cdot f'(x)$.